

CNC-X100 Diagnostic & Troubleshooting Guide

Product: CNC Machine | Model: CNC-X100 | Document Type: TROUBLESHOOTING

1. Diagnostic Overview & Alarm System

The CNC-X100 control unit monitors telemetry sensors continuously. When abnormal operating conditions occur, an alarm code is generated on the touchscreen HMI and logged into the system diagnostics table.

2. Error Code Diagnostic & Resolution Procedures

Error Code E105 - Cooling System Malfunction

Description: Coolant flow sensor detects flow rate below 8.0 L/min, or coolant pump thermal overload relay Q4 has tripped.

- Symptoms: Alarm banner on HMI, coolant pump stops, spindle power output limited, high temperature warning.
- Possible Causes: Low coolant level in 200L reservoir, clogged intake screen basket, damaged pump impeller, kinked delivery hose, or tripped Q4 circuit breaker.
- Diagnostic Checks: Inspect coolant sight glass on reservoir tank; check flow meter FM-1 output on diagnostic screen; inspect circuit breaker Q4 in electrical cabinet.
- Recommended Corrective Actions: 1. Top up coolant emulsion to 80% level. 2. Remove intake screen filter basket and flush debris with compressed air. 3. Reset thermal relay Q4. 4. Replace pump motor if winding resistance is unbalanced.

WARNING / SAFETY PRECAUTION:

SAFETY: Turn OFF electrical power before opening cabinet or servicing coolant pump. Wear chemical-resistant safety gloves and safety goggles.

Error Code E210 - Spindle Overload Warning

Description: Main spindle motor current exceeds 120% of rated continuous output for greater than 5 consecutive seconds.

- Symptoms: Spindle speed drops under heavy load, chatter marks on workpiece, E210 alarm displays on screen.
- Possible Causes: Excessive depth of cut, dull or chipped cutting inserts, hard spots in workpiece material, spindle bearing mechanical binding, or incorrect feed rate.
- Diagnostic Checks: Check spindle load meter on HMI; inspect cutting tool for chip welding or edge wear; manually rotate spindle shaft (power OFF) to feel for mechanical drag.
- Recommended Corrective Actions: 1. Press Feed Hold and inspect cutting tool. 2. Replace worn inserts or tool assembly. 3. Reduce axial depth of cut by

25%. 4. Increase spindle speed by 15% or reduce feed rate.

WARNING / SAFETY PRECAUTION:

SAFETY: Ensure spindle has come to a COMPLETE STOP before opening enclosure door. Do not touch hot cutting tools with bare hands.

Error Code E315 - Temperature Sensor Fault

Description: Thermal compensation sensor RTD-1 (Spindle Housing) or RTD-2 (Z-Axis Nut) signal out of valid range (-20°C to +120°C).

- Symptoms: Incorrect temperature readout on screen, thermal compensation disabled, E315 alarm active.
- Possible Causes: Loose wiring connector at CN7 board, severed thermistor wire, defective PT100 sensor probe, or strong electromagnetic interference.
- Diagnostic Checks: Inspect terminal block CN7 on I/O board; measure resistance of PT100 probe (nominal 108.4 Ohms at 21°C); check cable shield ground continuity.
- Recommended Corrective Actions: 1. Re-seat connector CN7 securely. 2. Replace defective temperature sensor probe assembly (Part # X100-SEN-315). 3. Repair loose ground shield wire.

WARNING / SAFETY PRECAUTION:

SAFETY: Disconnect main power breaker prior to probing electrical terminal blocks or measuring wiring resistance.

Error Code E420 - Lubrication System Pressure Warning

Description: Lubrication pressure switch PS-2 fails to detect 15 bar pressure within 30 seconds of automatic lube pump cycle start.

- Symptoms: Dry linear guide ways, lube pump runs continuously, E420 alarm active.
- Possible Causes: Empty slide-way oil tank, air trapped in lube supply lines, damaged distributor metering valve, clogged 10-micron oil filter, or faulty pressure switch PS-2.
- Diagnostic Checks: Check ISO VG 68 oil reservoir level; press manual lube prime button and observe pressure gauge; check linear guide ways for oil film.
- Recommended Corrective Actions: 1. Fill lube reservoir with clean ISO VG 68 slide-way oil. 2. Loosen air bleed screw on pump manifold until bubble-free oil flows. 3. Replace clogged oil filter element (Part # X100-LUB-010).

WARNING / SAFETY PRECAUTION:

SAFETY: Immediately clean up any spilled slide-way oil on floor around machine to prevent slipping hazards.

3. Additional Machine Symptom Diagnostics

- Pneumatic Pressure Low (<6.0 bar): Verify shop supply air pressure; inspect pneumatic FRL moisture drain; adjust regulator knob to 6.5 bar.
- Spindle Vibration / Overheating: Inspect spindle oil chiller unit; clean

cooling fins; check BT40 tool holder pull stud torque (45 Nm).

- Safety Door Interlock Lockout: Inspect door proximity switch alignment; clear chips from door track; verify 24V DC signal at door switch.
- Emergency Stop Recovery: Twist red mushroom button clockwise to reset; press Power On button on HMI; home all 3 axes.